Engineering Mathematics - Matrices Cheat Sheet
Rank: number of linearly independent rows (or columns).
Determinant of 2x2 [[a,b],[c,d]] = ad - bc.
Inverse exists only if det(A) != 0.
Eigenvalues: solve det(A - lambda*I) = 0.
Cayley-Hamilton: every square matrix satisfies its own characteristic equation.
Sum of eigenvalues = trace(A); product of eigenvalues = det(A).